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ENHANCING DOPPLER AND FMCW RADARS VIA UNLIMITED SENSING

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ABSTRACT

The Unlimited Sensing Framework (USF) is a paradigm that breaks the dynamic range–resolution trade-off in conventional analog-to-digital converters by combining modulo sampling with advanced signal processing algorithms. In this work, we focus on USF-enabled radar systems. We investigate radar models that admit a sum-of-sines (SoS) structure which can be effectively exploited. We propose a single-channel spectral estimation method that integrates high-order time-domain differences with the Fourier-domain sparsity of the SoS model. This yields an extension of the classical Fourier–Prony algorithm that operates at reduced sampling rates while achieving substantial dynamic range gains. Hardware experiments with USF-enabled radars validate the theoretical developments, demonstrating the effectiveness of the proposed framework for both Doppler and FMCW modalities.

***Index Terms*—** Modulo analog-to-digital conversion, radar, spectral estimation, super-resolution, unlimited sensing framework.

1. INTRODUCTION

Towards USF-Radars. Radar is a prevalent and classical application in signal processing [1]. Its digitization relies on Shannon's sampling theorem [2], which in practice is implemented using analog-to-digital converters (ADCs). ADCs perform quantization in both time and amplitude. However, as power consumption and circuit complexity increase rapidly with the quantization bit budget [3], low-resolution sampling techniques for radar have been extensively investigated over the past two decades. Notable approaches include one-bit sigma–delta converters [4] and one-bit quantization schemes [5]. Despite these advances, practical radar systems remain constrained by the fundamental *dynamic range–resolution* trade-off inherent to conventional ADCs. USF-enabled radars (see Fig. 1) have recently emerged as a novel solution to the long-standing dynamic range–resolution trade-off, achieved by deliberately introducing modulo non-linearity into the sampling pipeline [6–8]. Since the first demonstration in [9], they have shown promising results, including (i) a 10 dB reduction in the quantization noise floor enabling reliable weak target detection [10], and (ii) marked gains in signal and gesture classification accuracy [11]. These advances underscore the need for continued exploration and seamless integration of modulo ADC hardware within next-generation radar pipelines.

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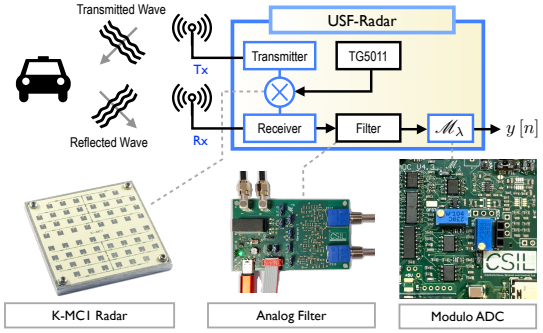


Fig. 1: USF-enabled Radar Setup. We use a K-MC1 transceiver with carrier frequency of 24.125 GHz for both Doppler (see Fig. 3) and FMCW (see Fig. 4) configurations. The analog output of the radar is filtered and amplified by a custom-designed circuit and subsequently fed into the $\mathscr{M}_\lambda$–ADC [8, 19].

Among various radar signal models, the sum-of-sines (SoS) representation is fundamental to both Doppler and FMCW configurations [1]. Within this setting, the recovery problem arising from USF-enabled Radar acquisition can be formulated as USF-SE (USF-based spectral estimation). Since its inception in [12], USF-SE has been investigated along several directions: (i) algorithm design for recovering weak–strong mixtures [13, 14], (ii) derivation of Cramér–Rao bounds [15, 16], and (iii) development of multi-channel architectures operating at sub-Nyquist rates [17, 18]. Despite this progress, state-of-the-art *single-channel* USF-SE methods remain limited in terms of input bandwidth, sampling rate, and dynamic range (DR) extension, preventing USF-enabled Radar from reaching its full potential. Furthermore, existing USF-enabled Radar approaches often treat SoS as a *special case* of bandlimited signals, overlooking its inherent Fourier-domain sparsity and thus failing to fully exploit model priors in practical scenarios. This work is driven by the gap between the demanding requirements of real-world applications and the constraints of current USF-SE algorithms.

Challenges and Related Work. The mainstream recovery methods in USF-enabled Radar literature [9, 10] is based on the Unlimited Sampling algorithm (US-Alg) [6, 7], utilizing the high-order difference (HoD) operation to gain high DR reconstruction. This has led to the first hardware validation on Doppler radar setup with $4\times$ DR gain under $9\times$ oversampling factor (compared to the peak frequency of 450 Hz) [10]. In the USF-SE literature, the estimation methods can be broadly categorized into two groups: (i) approaches designed for bandlimited signals, such as time-domain methods, including linear prediction [20], unlimited one-bit [21], WaveBus [22] and iterative signal sieving [23, 24]; Fourier-domain methods such as the "Fourier-Prony" algorithm [8] and its variants [13, 25]. Such kind of methods take SoS as bandlimited signals for reconstruction and then

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perform spectral estimation; (ii) approaches designed for SoS, such as HoD approach [12, 14] and optimization approach [13]. In [13], the authors reformulated the spectral fitting in [8] as sparse recovery and developed a Ioth's-type condition to upper bound the sample variation. This naturally compromises the DR compressibility, resulting in $5\times$ DR gain under $10\times$ OF (oversampling factor) using synthesized modulo data under the FMCW radar setup.

Novel Approach: Time-Frequency Hybridization. The US-Alg exploits the commutativity of modulo and difference operators (Eq. (16), [7]), enabling arbitrary HDR recovery with a sampling rate independent of input DR. In contrast, the FP-Alg models modulo-induced folds as sinusoidal components in the Fourier domain (Eq. (18), [8]), achieving tighter sampling rates but with DR dependence. In our upcoming work [26], we introduce a hybrid of these methods, the HoD-FP algorithm, which jointly leverages time-domain commutativity and Fourier-domain parametric structure to improve both OF and DR extension. While originally developed for bandlimited signals, we here extend the hybridization strategy to spectral estimation. The key insight is that commutativity can be used to sparsify the effective number of unknown folds, which can then be estimated robustly. Extending from bandlimited priors to the SoS structure, this time–frequency approach achieves estimation resolution beyond the capability of HoD-FP (see Section 4).

Contributions. In this paper, we propose a novel, single-channel USF-SE method, combining the complementary strengths of time- and Fourier-domain methods, translating into clear performance enhancement in USF-enabled Radar setups over existing approaches.

❑ **Algorithm.** We propose a novel USF-based spectral estimation method, HoD-SE, which is noise resilient (see Fig. 2) and operates at low sampling rates where existing methods fail (see Fig. 3-Fig. 4).

❑ **Experiments.** We implement the USF-enabled Radar pipeline as our testbed and conduct hardware experiments for both Doppler and FMCW configurations under challenging scenarios—(i) $12.35\times$ DR extension and (ii) $3.45\times$ oversampling. We demonstrate the practical advantages of our method by benchmarking its performance against existing approaches.

2. HIGH-ORDER SPECTRAL ESTIMATION

Problem Formulation. The radar signals utilized in this paper can be characterized as a sum of sinusoids, defined as,

$$g(t)=\sum\nolimits_{k=0}^{K-1} g_k e^{\jmath\omega_k t},\quad g\in\mathbf{L}_\infty(\mathbb{R}). \tag{1}$$

The non-linear folding operation maps $g(t)$ into a continuous-time, low-dynamic-range signal $y(t)=\mathscr{M}_\lambda(g(t))$, where,

$$\mathscr{M}_\lambda : g\mapsto 2\lambda\left(\left[\!\left[\frac{g}{2\lambda}+\frac{1}{2}\right]\!\right]-\frac{1}{2}\right),\ \begin{array}{l}[\![g]\!]\overset{\text{def}}{=}g-\lfloor g\rfloor\\ \lambda>0\end{array} \tag{2}$$

and $\lfloor g\rfloor=\sup\{k\in\mathbb{Z}|\,k\leqslant g\}$ denotes the residue of g. Subsequently, $y(t)$ is pointwise sampled, $y[n]=\mathscr{M}_\lambda(g(t))|_{t=nT}$, $n\in\mathbb{I}_N$, where T is the sampling step and N is the number of samples ($\mathbb{I}_N=\{0,\dots N-1\}$). In practice, since the modulo-folding occurs in analog-domain, the noisy folded measurements can be expressed as [16, 18], $y_w[n]=y[n]+w[n]$, where w is a mixture of (i) thermal noise [27] and (ii) quantization noise. Given $\{y_w[n]\}_{n\in\mathbb{I}_N}$, our goal is to develop a spectral estimation method that offers DR compressibility, noise resilience and high-resolution.

Integration of Modulo and Partitioning in HoD Domain. The key insight of HoD-SE approach lies in *introducing modulo operation* in high-order difference domain, promoting the *sparsity* of the resulting residue. To show this, we start with modular decomposition [7],

$$g=\mathscr{M}_\lambda(g)+\varepsilon_g,\quad \varepsilon_g(t)=\sum\nolimits_{m=1}^{M} c_m u(t-\tau_m) \tag{3}$$

where $u(\cdot)$ is the unit step function, and $c_m\in 2\lambda\mathbb{Z}$ and $\tau_m\in\mathbb{R}$ denote the fold amplitude and instant. Let $\{\Delta^{(h)}g,\Delta^{(h)}y,\Delta^{(h)}\varepsilon_g\}$ denote the h-th order finite difference of $\{g,y,\varepsilon_g\}$, respectively. Performing the h-th order finite difference on $\{y[n]\}_{n\in\mathbb{I}_N}$ leads to,

$$\Delta^{(h)}g[n]=\Delta^{(h)}y[n]+\Delta^{(h)}\varepsilon_g[n] \tag{4}$$

where $\Delta^{(h)}\varepsilon_g[n]=\sum_{m=1}^{M_h} c_{m,h}\delta[n-n_{m,h}]$, $c_{m,h}\in 2\lambda\mathbb{Z}$, $n_{m,h}\in\mathbb{I}_{N-h}$, $h\in\mathbb{Z}^+$ and $M_h\geqslant M_{h-1}, M_1=M$. In the absence of noise, $\|\Delta^{(h)}g\|_{\ell_\infty}$ shrinks exponentially as h increases, provided that $T\leqslant(2e\max_k|\omega_k|)^{-1}$ [7]. This leads to constant-factor oversampling, allowing for accessing $\Delta^{(h)}g$ from $\Delta^{(h)}y$. However, the presence of noise w deteriorates the *commutativity* needed for recovery. To enhance noise resilience while preserving DR compressibility, in this paper, we incorporate modulo non-linearity to promote the sparsity of the corresponding residue. Specifically, performing modular decomposition on $\Delta^{(h)}g$ leads to,

$$\Delta^{(h)}g[n]=\mathscr{M}_\lambda(\Delta^{(h)}g[n])+\varepsilon_{\Delta^{(h)}g}[n] \tag{5}$$

where $\varepsilon_{\Delta^{(h)}g}$ is the residue of $\Delta^{(h)}g$. By definition, we have that,

$$\Delta^{(h)}\varepsilon_g\in 2\lambda\mathbb{Z}\implies\mathscr{M}_\lambda(\Delta^{(h)}g[n])=\mathscr{M}_\lambda(\Delta^{(h)}y[n])$$

and therefore, (5) translates to,

$$\begin{array}{l}\mathscr{M}_\lambda(\Delta^{(h)}y[n])=\Delta^{(h)}g[n]-\varepsilon_{\Delta^{(h)}g}[n],\\ \varepsilon_{\Delta^{(h)}g}[n]=\sum_{m=1}^{\overline{M}_h}\overline{c}_{m,h}\delta[n-\overline{n}_{m,h}]\end{array} \tag{6}$$

where $\overline{c}_{m,h}\in 2\lambda\mathbb{Z}$ and $\overline{n}_{m,h}\in\mathbb{I}_{N-h}$. Note that, the cardinality of $\varepsilon_{\Delta^{(h)}g}$ reduces as h increases, *i.e.* $\overline{M}_{h-1}\geqslant\overline{M}_h$. In summary, integrating the modulo non-linearity with HoD enables a sparse representation of $\varepsilon_{\Delta^{(h)}g}$. In this sense, our method in this paper generalizes the "sparsity concept of the residue" [8,23], relaxing the burden on oversampling. Furthermore, from (1), $\Delta^{(h)}g$ can be expressed as,

$$\Delta^{(h)}g[n]=\sum\nolimits_{k=0}^{K-1} g_{k,h}e^{\jmath\omega_k Tn},\ g_{k,h}=g_k(e^{\jmath\omega_k T}-1)^h \tag{7}$$

which is a sum of sinusoids with the same frequency components. Hence, $\mathscr{M}_\lambda(\Delta^{(h)}y[n])$ can be re-characterized as,

$$\mathscr{M}_\lambda(\Delta^{(h)}y[n])=\sum_{k=0}^{K-1} g_{k,h}e^{\jmath\omega_k Tn}-\sum_{m=1}^{\overline{M}_h}\overline{c}_{m,h}\delta[n-\overline{n}_{m,h}] \tag{8}$$

which gives rise to a *low-dimensional, parametric* model. Leveraging (8), in real-world scenarios, the USF-based spectral estimation problem can thus be posed as,

$$\begin{array}{l}\min\limits_{\substack{\mathbf{g}_h,\boldsymbol{\omega},\\ \overline{\mathbf{c}}_h,\overline{\mathbf{n}}_h}}\ \sum_{n=0}^{N-h-1}\left|\mathscr{M}_\lambda(\Delta^{(h)}y_w[n])-\Delta^{(h)}g[n]+\varepsilon_{\Delta^{(h)}g}[n]\right|^2\\ \text{s.t. }\ \varepsilon_{\Delta^{(h)}g}[n]=\sum_{m=1}^{\overline{M}_h}\overline{c}_{m,h}\delta[n-\overline{n}_{m,h}],\ \begin{array}{l}\overline{c}_{m,h}\in 2\lambda\mathbb{Z}\\ \overline{n}_{m,h}\in\mathbb{I}_{N-h}\end{array}\\ \qquad\Delta^{(h)}g[n]=\sum_{k=0}^{K-1} g_{k,h}e^{\jmath\omega_k Tn},\ \ g\in\mathbf{L}_\infty(\mathbb{R}).\end{array} \tag{9}$$

(9) is a *dual-super-resolution* (SRes) problem that exhibits the same

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"sinusoids + spikes" structure in both time- and Fourier-domain. Solving the minimization in (9) is non-trivial, necessitating the investigation of two key questions: i) How to efficiently find the spectral parameters $\{\mathbf{g}, \boldsymbol{\omega}\}$?, and ii) How to select the difference order h? We will address these two questions in the following content.

Spectral Estimation via Fourier-Partitioning. Given the complex constraints, an effective strategy is to decompose (9) into two tractable sub-problems, which can be tackled utilizing an alternating minimization approach [23]. However, the low-dimensional, parametric structure of $\Delta^{(h)}g$ enables the development of an efficient, non-iterative approach to retrieve the spectral parameters $\{\mathbf{g}, \boldsymbol{\omega}\}$ directly in the HoD domain, eliminating the need for signal unfolding.

Denote by $\{\widehat{g}_h, \widehat{y}_h, \widehat{\varepsilon}_{g,h}\}$ the Discrete Fourier Transform (DFT) of $\{\Delta^{(h)}g, \Delta^{(h)}y, \varepsilon_{\Delta^{(h)}g}\}$. By definition, we have,

$$\begin{aligned}\widehat{g}_h[l] &= \sum\nolimits_{n=0}^{N-h-1} \Delta^{(h)} g[n]\, e^{-\jmath\frac{2l\pi n}{N-h}} \\ &= \sum\nolimits_{k=0}^{K-1} g_{k,h} \tfrac{1-e^{\jmath\omega_k(N-h)T}}{1-e^{\jmath(\omega_k T-\frac{2l\pi}{N-h})}},\quad l\in\mathbb{I}_{N-h}. \end{aligned} \tag{10}$$

As a result, in the out-of-band region, $[L_{g,h}, N-h-L_{g,h}]$, $L_{g,h} = \lceil 2\max_k|\omega_k|\,T(N-h)/2\pi\rceil$, we can derive that,

$$|\widehat{g}_h[l]| \leqslant \sum\nolimits_{k=0}^{K-1} |g_{k,h}|\, \tfrac{|\sin(\omega_k(N-h)T/2)|}{\min_k \sin\left((\frac{2L_{g,h}\,\pi}{N-h}-\omega_k T)/2\right)} \tag{11}$$

where $g_{k,h}$ decreaese exponentially regarding h under OF >3. Thus, under appropriate oversampling, $|\widehat{g}_h[l]|$ is upper bounded in the out-of-band region, resulting in the Fourier-separation,

$$\widehat{y}_h[l] \approx -\widehat{\varepsilon}_{g,h}[l] = \sum\nolimits_{m=1}^{\overline{M}_h} \overline{c}_{m,h} e^{\jmath u_{m,h} l},\ \ \overline{c}_{m,h}\in 2\lambda\mathbb{Z} \tag{12}$$

where $u_{m,h} = \frac{-2\pi\overline{n}_{m,h}}{N-h}$. Thus, $\{\overline{c}_{m,h}, \overline{n}_{m,h}\}_{m\in\mathbb{I}_{\overline{M}_h}}$ can be estimated by solving the following minimization,

$$\begin{aligned}&\min_{\overline{\mathbf{c}}_h,\overline{\mathbf{n}}_h}\ \sum\nolimits_{l=L_{g,h}}^{N-h-L_{g,h}} \left|\widehat{y}_{h,w}[l] - \sum\nolimits_{m=1}^{\overline{M}_h} \overline{c}_{m,h} e^{\jmath u_{m,h} l}\right|^2 \\ &\text{s.t.}\quad \overline{c}_{m,h}\in 2\lambda\mathbb{Z},\ \ \tfrac{u_{m,h}(N-h)}{-2\pi}\in\mathbb{I}_{N-h}. \end{aligned} \tag{13}$$

(13) is a typical frequency estimation problem, which can be addressed using high-resolution techniques, such as matrix pencil [28] and atomic norm minimization [29]. In this paper, we use the matrix pencil approach [28] due to its low computational complexity.

Having $\{\overline{\mathbf{c}}_h, \overline{\mathbf{n}}_h\}$ retrieved, we can reconstruct $\varepsilon_{\Delta^{(h)}g}$ via (6). Then, ε_g can be reconstructed (up to $2\lambda\mathbb{Z}$) by iteratively performing the anti-difference operator (see [7] Theorem 2), and g can be recovered via $\widetilde{\varepsilon}_g + y_w \mapsto \widetilde{g}$. With the sinusoidal model (1), the spectral parameters $\{\mathbf{g}, \boldsymbol{\omega}\}$ can then be estimated by addressing[1],

$$\min_{\mathbf{g},\boldsymbol{\omega}}\ \sum\nolimits_{n=0}^{N-1} \left|\widetilde{g}[n] - \sum\nolimits_{k=0}^{K} g_k e^{\jmath\omega_k T n}\right|^2 \tag{14}$$

which can be attained by using matrix pencil approach again. The algorithmic procedure is summarized in Algorithm 1.

Adaptive Difference Order Estimation. The finite difference depth h is a key parameter for our approach, controlling the noise level as $\mathbb{E}((\Delta^{(h)}w)^2) = \frac{(2h)!}{(h!)^2}\sigma^2$. The optimal choice of h can be found by minimizing the reconstructed mean-square error (MSE) on g,

$$\min_{h\in\mathbb{Z}^+}\ \sum\nolimits_{n=0}^{N-1} |g[n] - \theta(y_w)[n]|^2 \tag{15}$$

where $\theta(\cdot)$ is the estimator, which refers to the HoD-SE method in our context. However, solving (15) requires prior information of g, which is impractical in real-world scenarios. To this end, we relax (15) by minimizing the data consistency,

$$\begin{aligned}&\min_{h\in\mathbb{I}_{\overline{h}}}\ \sum\nolimits_{n=0}^{N-1} |y_w[n] - \mathscr{M}_\lambda(\theta(y_w)[n])|^2 \\ &\text{where}\quad \overline{h} = \sup\{h\in\mathbb{Z}^+ \mid (\tfrac{(2h)!}{(h!)^2})^{1/2} \leqslant \tfrac{\lambda}{\sigma}\}. \end{aligned} \tag{16}$$

$\overline{h}$ is the maximum difference order depending on λ and σ. Hence, the optimal difference order can be efficiently found by solving (16).

HoD-SE: High-Order Spectral Estimation.

Input: Noisy folded measurements $\{y_w[n]\}_{n\in\mathbb{I}_N}$ and h.
1: Conduct h-th order difference on y_w.
2: Calculate $\widehat{y}_{h,w}[l]$ via DFT.
3: Solve (13) using matrix pencil [28].
4: Reconstruct $\{\varepsilon_{\Delta^{(h)}g}, \Delta^{(h)}g\}$ via (6).
5: Recover $\widetilde{\varepsilon}_g$ via iterative anti-difference [7].
6: Reconstruct $\widetilde{g}$ via $\widetilde{\varepsilon}_g + y_w \mapsto \widetilde{g}$.
7: Solve (14) using matrix pencil [28].
8: Reconstruct g via (1).
Output: The recovered signal g and $\{\mathbf{g}, \boldsymbol{\omega}\}$.

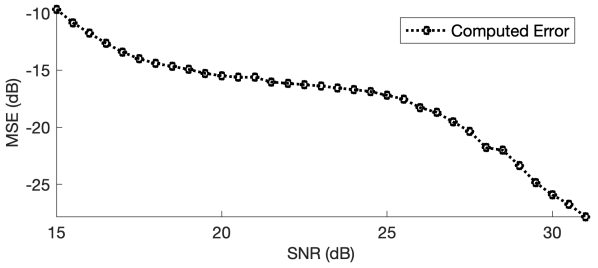


Fig. 2: Performance Evaluation across Varying Noise Levels. The input signal entails 3 sinusoids with frequencies $\{341.81, 498.69, 512.75\}$Hz. The noisy modulo samples are generated with 10-fold DR extension and 9.75-fold oversampling. Each SNR sample is averaged over 5000 realizations.

3. NUMERICAL EXPERIMENTS

In order to assess the robustness of the proposed method, we conduct the following numerical experiment: the real-values input g entails three sinusoids with frequencies of $\{341.81, 498.69, 512.75\}$Hz. We set $\lambda = 0.20$ and $T = 100$ ms, which results in 10-fold dynamic range extension and 9.75-fold oversampling. We generate the modulo samples via (2) and add white Gaussian noise, spanning noise level from 15 dB to 31 dB. The result at each SNR (signal-to-noise ratio) sample is averaged over 5000 random realizations to ensure statistical reliability. From the performance curve plotted in Fig. 2, we can deduce that: 1) In moderate noise regime, *i.e.* $[17, 25]$ dB, our HoD-SE algorithm attains stable performance, and 2) In low noise regime, *i.e.* $\geqslant 26$ dB, our HoD-SE method consistently delivers 1.87 dB enhancement in MSE per dB in SNR.

[1]We increase the model order by 1 to compensate for the constant drift.

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Table 1: Hardware Experiments: Parameters and Performance Metrics.

Figure	Exp. No.	OF	T	$\|g\|_{\mathbf{L}_\infty}$	λ	N	M	h	f_k	$\widetilde{f}_k$	$\mathcal{E}(\widetilde{\mathbf{g}}, \mathbf{g})$						
			(μs)	(V)	(V)				(Hz)	(Hz)	HoD-SE	HoD-FP [26]	IterSiS [23]	FP-Alg [8]	WaveBus [22]	B2R2 [25]	US-Alg [7]
Fig. 3	I	6.67	9	4.20	0.34	400	173	2	[3576.0]	[3575.9]	$\mathbf{1.87 \times 10^{-2}}$	2.51×10^{0}	1.14×10^{1}	9.51×10^{0}	7.58×10^{0}	3.31×10^{1}	9.40×10^{0}
Fig. 4	II	3.45	576	2.68	0.47	400	215	4	[50.0, 100.0, 150.0, 199.9]	[49.9, 100.0, 150.0, 200.0]	$\mathbf{5.20 \times 10^{-4}}$	1.66×10^{0}	2.42×10^{0}	2.31×10^{0}	2.18×10^{0}	4.89×10^{0}	1.63×10^{1}
—	III	3.51	13	0.93	0.36	400	61	4	[7999.5, 9000.0, 9999.9]	[8000.5, 9000.0, 10000.8]	$\mathbf{2.60 \times 10^{-4}}$	1.27×10^{0}	4.92×10^{-2}	3.04×10^{0}	1.83×10^{-1}	2.34×10^{-1}	1.06×10^{0}
—	IV	3.57	324	2.32	0.47	400	203	4	[5.9, 200.0, 397.5, 402.1]	[5.9, 200.0, 397.4, 402.1]	$\mathbf{4.30 \times 10^{-4}}$	2.40×10^{2}	1.90×10^{0}	1.50×10^{0}	4.38×10^{0}	1.65×10^{1}	6.30×10^{1}
—	V	4.35	20	1.19	0.37	400	135	2	[3500.2, 4500.1, 5500.1]	[3501.0, 4500.0, 5500.3]	$\mathbf{3.70 \times 10^{-3}}$	4.46×10^{-1}	6.39×10^{-1}	1.12×10^{-1}	5.14×10^{-1}	2.54×10^{0}	5.43×10^{0}

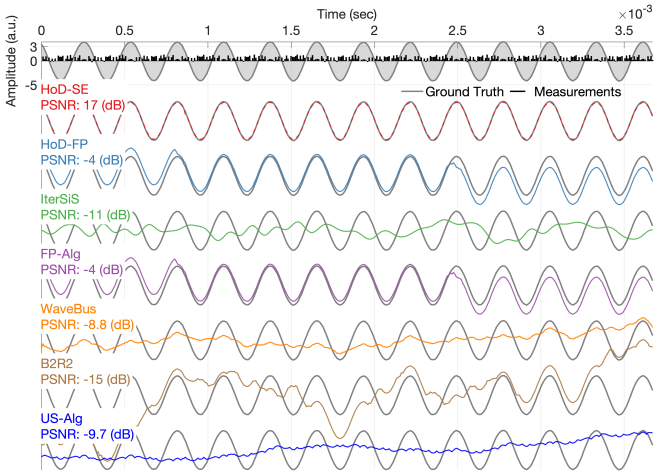


Fig. 3: Hardware Experiment: Doppler Radar. HoD-SE attains 12.35-fold DR extension and recovers the signal of 3.58 kHz with Hz-scale resolution. HoD-SE outperforms the existing methods with 21 dB enhancement.

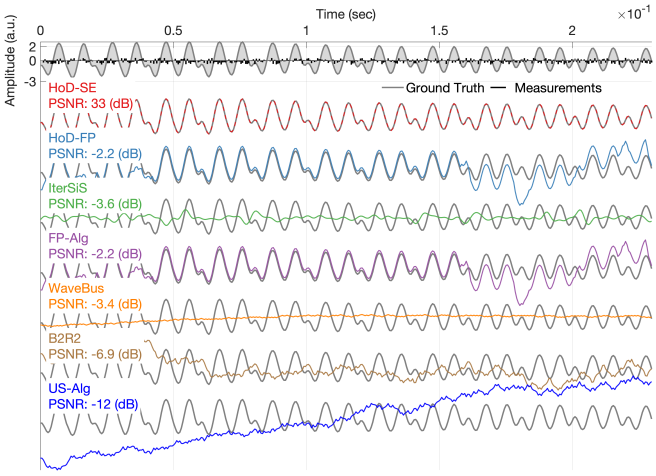


Fig. 4: Hardware Experiment: FMCW Radar. We push the OF down to 3.45, yielding $M = 215$ folds (3) where most of the recovery methods cannot handle. Despite the low OF and measurement noise, HoD-SE super-resolves 4 sinusoids with frequency estimation error within 1 Hz and PSNR of 33 dB.

4. HARDWARE EXPERIMENTS

Beyond numerical experiment, we incorporate $\mathscr{M}_\lambda$–ADC [8, 19] with radar as the testbed (see Fig. 1) to validate our HoD-SE algorithm in real-world scenarios. The radar signal is characterized as a sum of sinusoids for which their frequencies convey the information of interest, such as *speed* with Doppler radar and *range* with FMCW radar [1]. We demonstrate the capabilities and advantages of HoD-SE algorithm by quantitatively pushing the key performance metrics, (i) DR (dynamic range) extension and (ii) OF (oversampling factor) and benchmark its recovery precision against several existing methods, including HoD-FP [26], IterSiS [23], FP-Alg [8], WaveBus [22], B2R2 [25] and US-Alg [7]. The experimental parameters and performance metrics are tabulated in Table 1.

Doppler Radar. In the radar experiments, we utilize the experimental setup as depicted in Fig. 1 with carrier frequency $f_{\mathsf{TX}} = 24.125$ GHz for both Doppler and FMCW configurations. We use PicoScope 3403D to digitize the ground truth and folded signal with 8 bits and 7 bits resolution, respectively. In Doppler setup, the speed of the moving target v_k can be deduced from frequencies of the radar signal via $v_k = \frac{c}{4\pi f_{\mathsf{TX}}}\omega_k$ where $c = 3 \times 10^8$ m/s is the speed of light. To emulate a moving target, we use a K-DT1 Target Simulator with speed of 80 ± 1 km/h. We set $\lambda = 0.34$ and $T = 9$ μs, leading to 12.35-fold DR extension and 6.67-fold oversampling. The large DR yields closely-spaced folds, causing suboptimal performance of existing recovery methods, as shown in Fig. 3 and Table 1. Here, our HoD-SE algorithm offers frequency estimation down to 1 Hz scale resolution, showcasing the HDR capability in real-world scenarios.

FMCW Radar. Under the FMCW configuration, the range information R_k can be deduced from the frequencies of the radar measurements via $R_k = \frac{cT_{\mathsf{M}}}{8\pi f_{\mathsf{M}}}\omega_k$ where $f_{\mathsf{M}} = 174$ MHz is the modulation depth and $T_{\mathsf{M}} = 20$ ms is the chirp period. The instantaneous frequency f_{TX} of our USF-enabled Radar scheme (see Fig. 1) is modulated using the function generator TG5011. We set $\lambda = 0.47$ and $T = 576$ μs, giving rise to 5.70-fold DR extension and 3.45-fold oversampling that approaches the theoretical minimal OF $= 3$ (see [30], Theorem 1). The combination of HDR and low OF results in dense folds $M = 215$ (3), where the mainstream recovery methods, such as IterSiS and FP-Alg, cannot handle since they require $M < \frac{N}{2} = 200$. Despite challenges of low OF and measurement noise, our HoD-SE method consistently delivers accurate signal recovery, outperforming the state-of-the-art approach with 35 dB improvement, as shown in Fig. 4. This translates into spectral estimation down to 0.1 Hz resolution, effectively showcasing the efficacy and practical utility of our HoD-SE method for high-resolution USF-enabled Radar applications.

5. CONCLUSION

The Unlimited Sensing Framework overcomes the trade-off between optimizing dynamic range and digital resolution within a fixed bit budget, which is intrinsically rooted in the conventional sampling paradigms. In this paper, we develop a time-frequency hybridized method for USF-based spectral estimation, enabling super-resolved spectral estimation at low sampling regimes where existing methods fail. We benchmark our method with state-of-the-art approaches and demonstrate its superior performance through hardware experiments on both Doppler and FMCW USF-enabled Radar configurations.

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